# Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by total mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (104 sites, 381 site-years) · deadtrees v4.1

**XGBoost (Optuna) · 12m** | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

**GPPsat (µmol m⁻² s⁻¹)**

62% improved | 60% improved | 64% improved | 62% improved

**NEPmax (µmol m⁻² s⁻¹)**

74% improved | 73% improved | 73% improved | 70% improved

**ETmax (mm d⁻¹)**

57% improved | 56% improved | 59% improved | 53% improved

**WUE (g C mm⁻¹)**

54% improved | 67% improved | 57% improved | 55% improved

**ΔRMSE = RMSE(with M) − RMSE(without M)**

**Sites (sorted by ΔRMSE within each panel)**

Total mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 12.4 %, 40 sites)
- Mid (12.4–25.0 %, 43 sites)
- High (> 25.0 %, 21 sites)